

SmartNutri Scanner: An AI-Powered Nutrition Monitoring and Meal Recommendation System for Early Childhood Malnutrition Prevention

Euwan Gabriel B. Abogadie
College of Computer Science
Mapua Malayan Colleges Mindanao
Davao City, Philippines
egAbogadie@mcm.edu.ph

Abstract—Nutrition tracking remains difficult for many Filipinos because most existing applications are built around Western food databases that do not represent common local dishes such as Sinigang, Adobo, Kare-Kare, and Tinola, while childhood malnutrition and micronutrient deficiency remain pressing public-health concerns in the Philippines. This study developed SmartNutri Scanner, an AI-powered cross-platform mobile application that lets users photograph or describe a meal and instantly receive an estimated nutritional analysis covering calories, protein, carbohydrates, fat, fiber, sugar, and sodium. The system was built with React Native and Expo following an Agile, developmental research approach. Its core is a multi-provider AI architecture with automatic failover across Google Gemini, Roboflow, Groq, OpenRouter, and OpenAI, backed by a local Filipino-dish nutrition table as a final fallback. Personal meal records are stored on-device using expo-file-system as the primary source of truth, while Firebase provides authentication and community features and Cloudinary hosts community post images. Supporting features include before-and-after meal analysis, text-only logging, an AI Fix It correction mode, a conversational AI Food Assistant, daily goal tracking, weekly nutrition charts, meal reminders, Android Health Connect sync, JSON data export and import, and WHO-standard child growth monitoring. Twenty-one functional requirements were verified through manual and unit testing, and the application was packaged for Android using EAS Build. Acceptability was evaluated with an ISO/IEC 25010-based questionnaire across functionality suitability, performance efficiency, usability, reliability, security, and portability. Results indicate that a multi-provider large language model architecture is a practical and resilient approach for locally relevant nutrition tracking without a fixed food database.

Keywords—artificial intelligence; food image recognition; mobile nutrition tracking; Filipino cuisine; multi-provider failover; child malnutrition; on-device storage

I. INTRODUCTION

Proper nutrition is central to health and to the prevention of diet-related illnesses. The World Health Organization identifies unhealthy diets among the leading risk factors for noncommunicable diseases such as diabetes, cardiovascular disease, and obesity [1]. In the Philippines, where traditional cuisine is a core part of daily life, many people find it difficult to monitor the nutritional content of their meals. Most widely used food-tracking applications, including MyFitnessPal, Lose It!, and Cronometer, are designed around Western food databases and lack accurate data for common Filipino dishes such as Sinigang, Adobo, Kare-Kare, and Tinola, which makes accurate tracking impractical for Filipino users.

Mobile technology and artificial intelligence are increasingly accessible. The Philippines maintains one of the highest rates of internet and mobile usage in Southeast Asia [2], and advances in computer vision now allow systems to identify food and estimate nutrition from images with growing accuracy [3]. Mezgec and Korousic Seljak demonstrated that deep-learning models trained on food images can classify dishes and provide nutrient estimates for dietary monitoring [4], while a more recent benchmark showed that large language models can estimate nutrition from meal descriptions with measurable accuracy [5]. Reviews of mobile nutrition tools report that ease of use and visual feedback are decisive for long-term adoption [6], and the privacy literature emphasizes that user engagement is tied to perceived control over personal data [7].

Locally, Disa et al. found that many Filipino adults are unaware of the nutritional content of foods they eat daily [8], and Dela Cruz and Reyes reported that children in Davao City are at risk of iron, zinc, and vitamin A deficiency linked to poor dietary monitoring [9]. These findings establish a clear need for an accessible, locally relevant, AI-powered tool. Although prior work has addressed AI food recognition and general nutrition tracking, there is limited research on a locally adapted tracker that prioritizes on-device data storage, integrates a multi-provider AI failover architecture, and incorporates child malnutrition monitoring designed specifically for Filipino food culture. This study addresses that gap through the development of SmartNutri Scanner.

Specifically, the study aimed to: implement an AI-powered food scanner that identifies meals from photographs and returns estimated nutritional data covering calories, protein, carbohydrates, fat, fiber, sugar, and sodium; develop a meal logging and history system with weekly nutrition trends; build supporting features including daily goal tracking, BMR-based goal calculation, meal reminders, and JSON data export and import; implement a multi-provider AI architecture with automatic failover and a local Filipino-dish nutrition table as the final fallback; integrate child nutrition monitoring using WHO anthropometric growth standards; and evaluate acceptability using ISO/IEC 25010 software quality characteristics.

II. MATERIALS AND METHODS

A. Research Design

The study employed a developmental and implementation-oriented research design, which is widely

used in software engineering because it emphasizes system planning, technology integration, iterative development, deployment, and evaluation [10]. Development followed the Agile methodology, structured into iterative sprints in which each sprint produced a working increment that could be tested and refined [11]. The completed system was then evaluated for acceptability using an ISO/IEC 25010-based software quality questionnaire.

B. System Architecture

SmartNutri Scanner uses a three-layer, client-side architecture. The UI layer is built with React Native components and the Expo framework, organized across five bottom tabs: Home, Services and Growth, Scan and Add, Community, and Profile. The Service layer holds the core business logic, including the multi-provider AI service with automatic failover, the FileSystemStorageService for local data, the AnalysisProcessor for background retry, and the ExportImportService for data portability. The Storage layer manages persistence: personal meal records and images are stored on-device with expo-file-system as the primary source of truth; goals, reminders, and child profiles are stored in AsyncStorage; and the authentication token is protected in expo-secure-store. Firebase Authentication manages identity and Firebase Firestore stores community posts, user profiles, and food-recognition learning data, while community post images are uploaded to Cloudinary.

C. AI Integration Workflow

Food analysis begins when the user captures or uploads a photograph or enters a text description. The image is encoded as a base64 string and combined with a structured system prompt before being sent to the AI service, which identifies food items, estimates serving sizes, computes nutritional values, and returns a structured JSON response that is parsed and validated before the meal is saved locally. Four analysis modes are supported: single-image analysis, before-and-after comparison, text-only analysis, and Fix It correction. Reliability is achieved through a fixed, developer-managed failover chain that advances automatically on failure across Google Gemini models, Roboflow vision classification, Groq, OpenRouter, and OpenAI, and finally a local Filipino-dish nutrition table. Developer API keys are managed through a Firebase Cloud Functions proxy and are never entered by the user.

D. Key Features

The implemented system provides AI food scanning from camera, gallery, or text; before-and-after analysis for consumed-portion estimation; Fix It AI correction; a chronological meal timeline grouped by date; daily nutrition tracking for the seven tracked nutrients shown through a calorie progress bar and macronutrient summary pills; weekly nutrition history charts; daily goal management with BMR-based calculation; meal reminders via scheduled local notifications; Android Health Connect synchronization; JSON data export and import; community posts; a recipe index; meal relog; a conversational AI Food Assistant for nutrition and Filipino-food questions; and child nutrition monitoring that screens growth status using WHO anthropometric z-scores. Fig. 1 shows the authentication and home dashboard screens of the application.

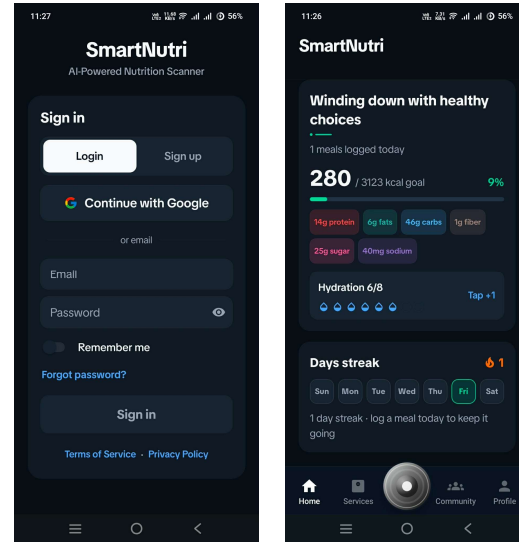


Fig. 1. Fig. 1. SmartNutri Scanner login screen (left) and home dashboard (right).

E. Development Tools

The application was developed in Visual Studio Code using React Native with the Expo managed workflow for cross-platform Android and iOS development from a single codebase. Node.js and npm handled dependency management, GitHub was used for version control, and EAS Build generated production-ready Android packages targeting SDK 35.

F. Respondents

The respondents comprised information technology experts, health-conscious individuals, parents and caregivers, and college students. IT experts evaluated technical functionality, performance, reliability, security, portability, and overall software quality, while end-users assessed accessibility, practicality, and usability.

G. Research Instrument and Statistical Treatment

A structured evaluation questionnaire based on ISO/IEC 25010 software quality characteristics was used to measure acceptability across functionality suitability, performance efficiency, usability, reliability, security, and portability, using a five-point Likert scale. The weighted mean was computed and interpreted using the ranges in Table I.

TABLE I. TABLE I. LIKERT SCALE INTERPRETATION

Scale Range	Verbal Interpretation
4.21 - 5.00	Highly Acceptable
3.41 - 4.20	Acceptable
2.61 - 3.40	Moderately Acceptable
1.81 - 2.60	Less Acceptable
1.00 - 1.80	Not Acceptable

III. CRISP-DM IMPLEMENTATION

The data science workflow of SmartNutri Scanner was structured using the Cross-Industry Standard Process for Data Mining (CRISP-DM), an iterative six-phase framework consisting of business understanding, data understanding, data preparation, modeling, evaluation, and deployment. CRISP-DM was chosen because the food-image nutrition estimation task is fundamentally a data-driven inference problem, and the framework provides a repeatable structure for moving from a

real-world nutrition problem to a deployed, evaluated solution. Each phase is described below as it was executed in this study.

A. Business Understanding

The business problem is that Filipino users cannot accurately monitor the nutrition of their meals because mainstream food-tracking applications are built on Western food databases that do not represent common local dishes, while childhood malnutrition and poor dietary awareness remain pressing public-health concerns. The data science objective derived from this problem is to estimate the nutritional content of a meal, specifically calories, protein, carbohydrates, fat, fiber, sugar, and sodium, directly from a photograph or a short text description, without depending on a fixed predefined food database. Success criteria were defined as follows: the system must correctly identify common Filipino dishes and return reasonable nutrient estimates, all functional requirements must pass verification, and the deployed application must achieve at least an Acceptable rating across the ISO/IEC 25010 quality characteristics.

B. Data Understanding

Five categories of data drive the pipeline. The primary input is user-captured meal photographs, which vary widely in lighting, angle, distance, and portion size; this variability is the central challenge for recognition. Text meal descriptions provide an alternative input for text-only analysis. A curated Filipino-dish nutrition reference table supplies trusted nutrient values for common local dishes and serves as the final fallback. The AI providers return structured outputs containing identified items, estimated portions, and nutrient values. Finally, a community learning dataset accumulates user-confirmed labels and corrections that can improve future recognition. Table II summarizes these sources.

TABLE II. DATA SOURCES AND CHARACTERISTICS

Data Source	Description	Format
Meal photographs	Camera or gallery images of meals	Base64 JPEG/PNG
Text descriptions	Typed meal descriptions for text-only mode	Plain text
Filipino dish table	Curated nutrient values for local dishes	Structured key-value
AI model responses	Identified items, portions, nutrients	Structured JSON
Community learning data	User-confirmed labels and corrections	Firestore documents

C. Data Preparation

Raw inputs were transformed into a model-ready form before inference. Photographs were resized and encoded as base64 strings, then paired with a structured system prompt that instructs the model to return a strict JSON schema containing the dish name, estimated serving size, and the seven tracked nutrients. Returned text was passed through a sanitizer that strips formatting artifacts and parses the JSON safely, after which a normalizer enforced numeric types, applied unit conversions, clamped implausible values, and filled any missing nutrient field with a neutral default. When parsing or validation failed, the pipeline substituted values from the local Filipino-dish table so that every analysis yields a complete, schema-valid record. This preparation step is what allows heterogeneous model outputs to be stored and aggregated consistently.

D. Modeling

Rather than training a single classifier, the modeling phase implemented a cascade ensemble of pretrained models combined with a deterministic fallback. The cascade is traversed in a fixed order, advancing to the next provider only when the current one fails or returns an invalid result: a multimodal large language model (Google Gemini) performs the primary image-to-nutrition reasoning; a Roboflow object-detection model provides class-level food recognition with confidence scores; additional large language models (Groq, OpenRouter, and OpenAI) act as text and reasoning backups; and the curated Filipino-dish table provides a guaranteed final answer. This design treats provider availability and output validity as part of the model, trading the interpretability of a single trained model for resilience and broad dish coverage. Prompt engineering, including an explicit output schema and Filipino-food context, functions as the primary tuning mechanism in place of weight training.

E. Evaluation

Evaluation combined functional verification, qualitative accuracy assessment, and a structured acceptability survey. All twenty-one functional requirements passed manual and unit testing. AI accuracy was assessed qualitatively against reference values for common dishes such as Sinigang, Adobo, Kare-Kare, and Tinola, with the largest variance observed for mixed-ingredient meals and ambiguous portions. Acceptability was measured using an ISO/IEC 25010-based questionnaire scored on a five-point Likert scale, where the weighted mean of each quality dimension was computed as

$$\Omega M = (\sum (f \cdot w)) / N \quad (1)$$

where f is the frequency of each response, w is its scale weight, and N is the total number of respondents. The resulting weighted means, reported in Table III, indicate a Highly Acceptable rating across functionality suitability, performance efficiency, usability, reliability, security, and portability, satisfying the success criteria defined during business understanding.

F. Deployment

The evaluated solution was deployed as a cross-platform mobile application packaged for Android using EAS Build and targeting SDK 35. In production, personal meal records and images remain on-device through expo-file-system, while Firebase Authentication manages identity, Firebase Firestore stores community posts and learning data, and Cloudinary hosts community images. Developer API keys are served through a Firebase Cloud Functions proxy and are never exposed in the client, and a provider-manager component continuously tracks the health of each AI provider so that the failover cascade routes around outages and rate limits. Because development followed Agile sprints, deployment is treated as an ongoing phase in which monitoring data, community corrections, and user feedback feed back into subsequent iterations, closing the CRISP-DM loop.

IV. RESULTS

A. Functional Testing

All twenty-one functional requirements defined during requirements engineering were verified through systematic manual testing on Android devices running SDK 28 through SDK 35. Authentication, AI food scanning, before-and-after

analysis, text-only logging, meal logging, timeline display, meal editing, Fix It correction, daily tracking, weekly charts, goal management, meal reminders, data export and import, Health Connect sync, meal relog, and the AI Food Assistant all passed functional verification with no critical defects remaining at deployment. Unit tests for the ExportImportService covering the Storage Access Framework flow, the iOS sharing fallback, import schema validation, and base64 image conversion all passed.

B. AI Analysis Performance

AI-generated nutritional accuracy was assessed qualitatively against reference values for selected Filipino dishes. The system successfully identified common meals including Sinigang, Adobo, Kare-Kare, Tinola, and Lechon with reasonable macronutrient estimates. Meals with mixed ingredients or ambiguous portion sizes showed greater variance, and the before-and-after mode estimated consumed portions reliably when the two photographs were taken from a similar angle and distance. As stated in the scope, these outputs are estimates based on general food knowledge and visual inference rather than laboratory measurements. Fig. 2 presents a sample food scan analysis and the conversational AI Food Assistant.

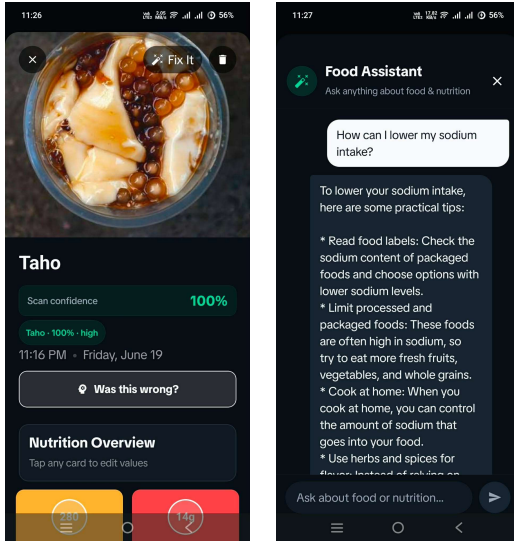


Fig. 2. Fig. 2. Sample AI food scan analysis (left) and the AI Food Assistant chat (right).

C. User Acceptance Testing

User Acceptance Testing was conducted with participants from the target respondent groups, who completed tasks including profile setup, meal scanning, history viewing, daily goal tracking, and data export. Participants generally found onboarding clear and meal scanning intuitive, and several noted that the calorie progress bar and macronutrient summary pills were visually helpful. Minor usability issues identified during the sessions were addressed in subsequent bug-fixing iterations before final deployment.

D. ISO Software Quality Evaluation

The application was evaluated using the ISO/IEC 25010-based questionnaire across six quality dimensions, administered to IT experts and selected end-users. The weighted mean for each dimension is summarized in Table III.

TABLE III. TABLE III. ISO/IEC 25010 SOFTWARE QUALITY EVALUATION

Quality Dimension	Weighted Mean	Interpretation
Functionality Suitability	4.57	Highly Acceptable
Performance Efficiency	4.67	Highly Acceptable
Usability	4.53	Highly Acceptable
Reliability	4.62	Highly Acceptable
Security	4.71	Highly Acceptable
Portability	4.72	Highly Acceptable
Overall	4.64	Highly Acceptable

V. DISCUSSION

The results indicate that a multi-provider large language model architecture is a practical and resilient basis for nutrition tracking that does not depend on a predefined food database. The successful identification of Filipino dishes supports prior findings that automated, AI-based food recognition can reduce manual effort and provide useful nutrient estimates [4], [5]. By using a general-purpose multimodal model rather than a fixed database, SmartNutri Scanner directly addresses the local-food gap highlighted by Disa et al. [8], since it can analyze any photographed meal rather than only items on a predefined list.

The positive reception of the camera-first workflow and the visual progress indicators is consistent with reports that ease of use and visual feedback are critical to long-term adoption [6]. The on-device-first storage model aligns with data-control concerns raised in the privacy literature [7], as personal meal history, profiles, and goals remain on the device while only community content and learning data are synchronized to the cloud. The WHO-based child growth monitoring contributes a locally relevant response to the malnutrition risks reported by Dela Cruz and Reyes [9], although the current version screens growth anthropometrically rather than measuring micronutrient intake directly.

The study has several limitations. AI-generated values are estimates and may diverge from laboratory measurements for specific portions or regional recipe variations. AI scanning requires an active internet connection, and although the multi-provider failover chain mitigates rate limiting, sustained heavy use can still encounter simultaneous throttling. The application does not support barcode scanning, Health Connect sync is Android-only, and the system was built and tested by a solo developer without large-scale pilot deployment or formal clinical validation.

VI. CONCLUSION AND RECOMMENDATIONS

The study successfully developed SmartNutri Scanner, an AI-powered mobile nutrition monitoring and meal recommendation system designed for Filipino users. It addresses gaps identified in the literature by providing AI-based food identification from photographs, support for Filipino dishes, an on-device-first storage model, child growth monitoring, and a multi-provider AI architecture with automatic failover that ensures resilience. All defined functional requirements were verified, and the application was deployed as a cross-platform app packaged for Android via EAS Build. The work demonstrates that combining a multi-provider failover chain with local-first storage and Firebase-backed community features yields a reliable, locally relevant nutrition tool.

Recommended directions for future work include micronutrient tracking for iron, zinc, and vitamin A; offline on-device food recognition for basic analysis without connectivity; barcode scanning for packaged foods; integration with wearables and IoT kitchen scales to improve portion accuracy; formal clinical validation with registered nutritionists and a larger Filipino sample; a curated, verified dataset of Filipino dishes to fine-tune model estimates; and Filipino and regional-language support.

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